The empirical recurrence for OEIS A197230 is false, and the correct one, of order 25

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Abstract

OEIS A197230 carries an empirical recurrence of order 22 contributed by R. H. Hardin. It is false. The entry publishes 22 terms and has offset 1, so the recurrence first says something at $n = 23$, and there it already fails: the true value is 1327965802062332 and the recurrence predicts 1327965802062198, a difference of 134. The sequence is a transfer-matrix count, so it is C-finite and its true minimal recurrence can be computed and proved; it has order 25, and its first 22 coefficients are exactly the published ones. The published line is therefore the correct recurrence with its last three terms dropped. Both statements — the failure and the corrected recurrence — are proved here in exact integer arithmetic.

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1 The sequence and the claim

OEIS A197230 is "Number of n X 3 0..4 arrays with each element x equal to the number its horizontal and vertical neighbors equal to 4,3,0,1,2 for x=0,1,2,3,4.". It has offset 1 and its published DATA is

$$2, 12, 55, 242, 1178, 5355, \ \ldots, \ 283580433219551$$

— 22 terms in all. The entry carries

> Empirical: a(n) = 2*a(n-1) +3*a(n-2) +38*a(n-3) +40*a(n-4) +10*a(n-5) -150*a(n-6) -375*a(n-7) -175*a(n-8) +637*a(n-9) +2013*a(n-10) +1368*a(n-11) -1614*a(n-12) -4361*a(n-13) -4043*a(n-14) +946*a(n-15) +4156*a(n-16) +3551*a(n-17) +552*a(n-18) -881*a(n-19) -629*a(n-20) -1033*a(n-21) +508*a(n-22).

As of the "Last modified" line on the live entry (Feb 07 2026 20:53:17, revision 11) the statement is recorded as empirical and nothing on the entry records it as settled or as corrected. Written out, the claim is

$$a(n) \;=\; 2\,a(n-1)+3\,a(n-2)+38\,a(n-3)+40\,a(n-4)+10\,a(n-5)-150\,a(n-6)-375\,a(n-7)-175\,a(n-8)+637\,a(\tag{1}$$

of order 22, with no range stated.

Note at once that (1) is untestable on the published terms: with offset 1 and order 22 the first index at which it asserts anything is $n = 23$, and the entry stops at $n = 22$.

2 The model

The condition is local to a cell and its four horizontal and vertical neighbours. Write $x_{t,u}$ for the entry in row t and column u, $1 \le u \le 3$, with values in $\{0, 1, 2, 3, 4\}$, and let

$$f = (4, 3, 0, 1, 2), \qquad \text{that is } f(0) = 4,\ f(1) = 3,\ f(2) = 0,\ f(3) = 1,\ f(4) = 2.$$

The entry asks that every cell satisfy

$$\#\big\{(d_1,d_2)\in\{(0,-1),(0,1),(-1,0),(1,0)\}:\ x_{t+d_1,\,u+d_2}=f(x_{t,u})\big\}\ =\ x_{t,u}, \tag{2}$$

positions outside the array not being counted. Because the neighbourhood reaches one row up and one row down, a state carrying a single row is not enough: take ordered pairs of consecutive rows as vertices, $|\mathcal{V}|=S=15625$, with $(p,c)\to(c,x)$ an edge when every cell of the middle row c satisfies (2) with p above and x below, and mark the pairs whose first row passes with nothing above, and those whose second row passes with nothing below. Then $a(n)=\mathbf{v}^\top N^{\,n-2}\mathbf{u}$ for $n\geq 2$.

This model reproduces the entry's DATA at all 22 published indices exactly. It was also checked against a direct enumeration of all 5^{3n} arrays for $n=1,2$, which gives 2 and 12, the entry's first two terms.

3 The published recurrence fails

Evaluating the model at $n=23$ gives

$$a(23)\ =\ 1327965802062332,$$

while the right-hand side of (1), applied to the 22 published terms, gives

$$1327965802062198.$$

The two differ by 134. Since (1) is asserted without any range and $n=23$ is the first index at which it asserts anything, the published recurrence is false. It continues to fail at $n=24,25,26,\ldots$; nothing about $n=23$ is exceptional.

4 The recurrence that does hold

Being a walk count, a is C-finite. Solving for a constant-coefficient recurrence over $\mathbb{Q}$ from the model's own terms, the least order that admits one is 25, and the coefficients come out integral:

$$a(n)\ =\ 2\,a(n-1)+3\,a(n-2)+38\,a(n-3)+40\,a(n-4)+10\,a(n-5)-150\,a(n-6)-375\,a(n-7)-175\,a(n-8)+637\,a(\tag{3}$$

The first 22 coefficients of (3) are, term for term, those of (1). The published line is (3) with its last three terms, $+28\,a(n-23)-82\,a(n-24)+40\,a(n-25)$, missing.

Theorem 1. (3) *holds for every $n>26$.*

Proof. Let $q(t)=t^{25}-\sum_i c_i t^{25-i}$ be its characteristic polynomial and $w=q(N)\mathbf{u}$. For $n\geq 27$,

$$a(n)-\sum_i c_i\,a(n-i)=\mathbf{v}^\top N^{\,n-27}w,$$

so the recurrence holds from there on if and only if $\mathbf{v}^\top N^m w=0$ for every $m\geq 0$; by the Cayley–Hamilton theorem N^S is an integer combination of $I,N,\ldots,N^{S-1}$, so checking $m<S$ suffices. The vector w was formed by 25 matrix–vector products in exact integer arithmetic and the run continued until $S+1$ consecutive zeros had been seen. Every value vanished from the index corresponding to $n=27$ onwards. ∎

Solving for the recurrence is only a way of PROPOSING (3); what makes it a theorem is the annihilation test above, which is independent of how the candidate was found.

5 Verification

Four checks.

First, the model reproduces all 22 published terms exactly, and its first two values were confirmed by brute-force enumeration over all 5^{3n} arrays.

Second, the failure at $n = 23$ was computed twice: once by iterating the transfer matrix, and once by applying (1) to the entry's own published integers with no matrices involved. Both are exact integer computations and they disagree by 134.

Third, (3) was checked directly on the model's terms for $n = 27, \dots, 120$ before any matrix argument was made, and holds at every one.

Fourth, the annihilation test was carried to the full Cayley–Hamilton bound rather than to a sampled prefix.

Remark 2. The shape of the error is worth stating plainly, because it decides what this paper is. The published coefficients are not wrong; they are incomplete. A recurrence fitted to a finite sample and then reported at an order the sample cannot support will look like this: correct as far as it goes, and false at the first index where it says anything. The correction is (3).

References

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